D4.2 Stability and change

NAME	DATE	TOTAL
		_____ / 42

About 40 minutes, or one section at a time. You will need a calculator. The number of marks shows how many separate points to make.

Section A: Recall

7 MARKS

1 **LIST** three requirements for ecosystem stability. [D4.2.2] [3]

2 **DEFINE** keystone species. [D4.2.4] [1]

3 **STATE** what BOD measures. [D4.2.7] [1]

4 **NAME** two pollutants that undergo biomagnification. [D4.2.8] [1]

5 **STATE** the size that separates microplastics from macroplastics. [D4.2.9] [1]

Section B: Understanding

15 MARKS

6 **EXPLAIN** how deforestation of the Amazon could lead to a tipping point. [D4.2.3] [4]

7 **OUTLINE** two factors that affect how sustainable a farm is. [D4.2.6] [2]

8 **EXPLAIN** how leaching of fertilizers can lead to the death of fish. [D4.2.7] [4]

9 **EXPLAIN** why plastic pollution harms marine animals. [D4.2.9] [3]

10 **OUTLINE** two methods used in rewilding. [D4.2.10] [2]

Section C: Working with data

14 MARKS

All data in this section are invented for practice. The patterns are realistic.

11 [D4.2.8]

TROPHIC LEVEL	MERCURY / MG PER G OF TISSUE
PLANKTON	0.01
SMALL FISH	0.10
TUNA	0.80
DOLPHIN	5.00

a **CALCULATE** how many times greater the concentration is in dolphins than in plankton. [1]

b **EXPLAIN** the trend. [3]

c **SUGGEST** why people are advised to limit the amount of large predatory fish that they eat. [1]

12 [D4.2.5]

YEAR	2000	2005	2010	2015	2020
STOCK OF A FISH / THOUSAND TONNES	800	620	440	300	360
CATCH / THOUSAND TONNES	200	190	180	40	40

a DESCRIBE the trend in stock. [2]

b DEDUCE, with a reason, whether the catch in 2000–2010 was sustainable. [2]

c EXPLAIN the change in stock after 2015. [2]

d STATE one further piece of data needed to assess this stock. [1]

e SUGGEST two management measures that could speed up recovery. [2]

Section D: Longer answer

6 MARKS

13 DISCUSS how human activities can threaten the stability of an ecosystem, and how they could be reduced. [D4.2.2–10] [6]